PassFrame
tuition

# Word Codes

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

**Worked examples — tutor talks through these first**

**A) Easy:** If MOUSE is coded as NPVTF (each letter shifts forward by 1), what does HORSE code as?
*Method & answer: Shift every letter of HORSE forward by 1: H→I, O→P, R→S, S→T, E→F. Answer: IPSTF.*

**B) Standard:** If the code for GARDEN is HBSEFO (each letter +1), what is the code for FLOWER?
*Method & answer: Shift every letter of FLOWER forward by 1: F→G, L→M, O→P, W→X, E→F, R→S. Answer: GMPXFS.*

**C) Advanced:** If CAT is coded as ZXQ (each letter shifts back by 3, wrapping round the alphabet), what is the code for DOG?
*Method & answer: Shift every letter of DOG back by 3, wrapping from A back to Z if needed: D→A, O→L, G→D. Answer: ALD.*

## Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** If the code for CAT is DBU (each letter shifts forward by 1), what is the code for DOG?

**2.** If the code for BIG is CJH (each letter shifts forward by 1), what is the code for BAD?

**3.** If the code for SUN is TVO (each letter shifts forward by 1), what is the code for SEA?

**4.** If the code for PEN is QFO (each letter shifts forward by 1), what is the code for PIG?

**5.** If the code for TOP is SNO (each letter shifts back by 1), what is the code for TIN?

**6.** If the code for RED is QDC (each letter shifts back by 1), what is the code for BOX?

**7.** If the code for CUP is EWR (each letter shifts forward by 2), what is the code for MAP?

**8.** If the code for BUS is DWU (each letter shifts forward by 2), what is the code for BAT?

**9.** In a code, every letter shifts forward by 1. The coded word is JDF. What is the original word?

**10.** If the code for FOX is GPY (each letter shifts forward by 1), what is the code for FAN?

**11.** If the code for TEN is RCL (each letter shifts back by 2), what is the code for PIG?

**12.** If the code for MUG is OWI (each letter shifts forward by 2), what is the code for BUN?

**13.** If the code for JAM is IZL (each letter shifts back by 1), what is the code for KID?

**14.** If the code for WEB is XFC (each letter shifts forward by 1), what is the code for WIN?

**15.** In a code, every letter shifts forward by 1. The coded word is HFU. What is the original word?

## Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** If the code for FALCON is IDOFRQ (each letter shifts forward by 3, wrapping round the alphabet where needed), what is the code for PARROT?

**2.** If the code for RABBIT is OXYYFQ (each letter shifts back by 3, wrapping round the alphabet where needed), what is the code for

**6.** A code reverses the order of the letters in a word. Using this code, GARDEN becomes NEDRAG. What does the same code turn TRAVEL into?

**7.** If the code for PURPLE is SXUSOH (each letter shifts forward by 3, wrapping round the alphabet where needed), what is the code for

**11.** Using a reverse-alphabet code (A ↔ Z, B ↔ Y, C ↔ X…), the word FOREST becomes ULIVHG. What does CASTLE become under the same code?

**12.** If the code for BRIDGE is XNEZCA (each letter shifts back by 4, wrapping round the alphabet where needed), what is the code for

MONKEY?

SILVER?

COUNTY?

**3.** If the code for WIZARD is ZLCDUG (each letter shifts forward by 3, wrapping round the alphabet where needed), what is the code for CASTLE?

**8.** In a code, every letter shifts back by 3, wrapping round the alphabet where needed. The coded word is DLIABK. What is the original word?

**13.** In a code, every letter shifts forward by 3, wrapping round the alphabet where needed. The coded word is ZLQGRZ. What is the original word?

**4.** In a code, every letter shifts forward by 4, wrapping round the alphabet where needed. The coded word is TIRGMP. What is the original word?

**9.** A code shifts consonants forward by 2 and vowels backward by 2, wrapping where needed. Using this rule, FROST becomes HTMUV. What does the same rule turn CRANE into?

**14.** If the code for STORM is XYTWR (each letter shifts forward by 5, wrapping round the alphabet where needed), what is the code for CANDLE?

**5.** If the code for SHADOW is UJCFQY (each letter shifts forward by 2, wrapping round the alphabet where needed), what is the code for MARBLE?

**10.** If the code for SPIDER is WTMHIV (each letter shifts forward by 4, wrapping round the alphabet where needed), what is the code for HAMMER?

**15.** If the code for TUNNEL is RSLLCJ (each letter shifts back by 2, wrapping round the alphabet where needed), what is the code for WHISKER?

## Standard / Hard

*— work through together, answer 5 in the lesson* *15 min (1 min per question)*

**1.** If HORSE is coded as JQTUG, what is the code for MOUSE? (Work out the shift from the first pair.)

**6.** A code shifts every letter forward by the number of letters in the word. Since CAT has 3 letters, each letter shifts by 3: CAT becomes FDW. Using the same rule, what does FIVE become?

**11.** We know BAT becomes EDW under a fixed forward shift code. What is the code for CAT under the same shift?

**2.** A code reverses a word's letters, then shifts each one forward by 1. Applying this to CAT gives UBD (reverse to TAC, then shift +1). What does the same rule turn LEMON into?

**7.** A code shifts consonants forward by 2 and vowels backward by 2, wrapping where needed. Using this rule, FISH becomes HGUJ. What does the same rule turn TRAIN into?

**12.** A code shifts every letter back by 4, wrapping from A round to Z. Using this code, what does the word DEAL become?

**3.** In a code, the 1st letter of a word shifts by 1, the 2nd by 2, the 3rd by 3, and so on. Using this rule, RIVER becomes SKYIW. What does the same rule turn HOUSE into?

**8.** A code first swaps each letter using the reverse-alphabet code (A ↔ Z, B ↔ Y…) and then shifts the result forward by 1. Applying this to DOG: reverse-code gives WLT, then shift +1 gives XMU. What does the same two-step rule turn CAT into?

**13.** A code swaps the 1st and last letters of a word, and shifts the middle letters forward by 1. Applying this to CHAT: swap to get THAC, then shift the middle letters to get TIBC. What does the same rule turn STOP into?

**4.** A code shifts every letter forward by 3, wrapping from Z back to A. Using this code, what does the word XYZ become?

**9.** In a code, letters in odd positions (1st, 3rd, 5th…) shift forward by 1, and letters in even positions (2nd, 4th…) shift backward by 1. Using this rule, LEMON becomes MDNNO. What does the same rule turn TIGER into?

**14.** A code doubles every letter (writes it twice) and then removes the vowels. Applying this to CAT: double to CCAATT, then remove vowels to get CCTT. What does the same rule turn DOG into?

**5.** If the code for CLOCK is ENQEM, what does the code CRRNG decode to? (Work out the shift first.)

**10.** A code shifts every letter forward by 1, then reverses the whole word. Applying this to CAT: shift to get DBU, then reverse to get UBD. Using the same rule, what does DOG become?

**15.** A code reverses the order of the letters in a word, then swaps each letter using the reverse-alphabet code (A ↔ Z, B ↔ Y, C ↔ X…). Applying this to CAT: reverse to get TAC, then swap each letter to get GZX. What does the same two-step rule turn DOG into?